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RESEARCH ARTICLE

# Application of StrucGP in medical immunology: site-specific *N*-glycoproteomic analysis of macrophages

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**Abstract** **The structure of *N*-glycans on specific proteins can regulate innate and adaptive immunity via sensing environmental signals. Meanwhile, the structural diversity of *N*-glycans poses analytical challenges that limit the exploration of specific glycosylation functions. In this work, we used THP-1-derived macrophages as examples to show the vast potential of a *N*-glycan structural interpretation tool StrucGP in *N*-glycoproteomic analysis. The intact glycopeptides of macrophages were enriched and analyzed using mass spectrometry (MS)-based glycoproteomic approaches, followed by the large-scale mapping of site-specific glycan structures via StrucGP. Results revealed that bisected GlcNAc, core fucosylated, and sialylated glycans (e.g., $HexNAc_4Hex_5Fuc_1Neu5Ac_1$, $N_4H_5F_1S_1$) were increased in M1 and M2 macrophages, especially in the latter. The findings indicated that these structures may be closely related to macrophage polarization. In addition, a high level of glycosylated PD-L1 was observed in M1 macrophages, and the LacNAc moiety was detected at Asn-192 and Asn-200 of PD-L1, and Asn-200 contained Lewis epitopes. The precision structural interpretation of site-specific glycans and subsequent intervention of target glycoproteins and related glycosyltransferases are of great value for the development of new diagnostic and therapeutic approaches for different diseases.**

**Keywords** macrophage; glycoproteome; glycopeptides; *N*-glycan structures; PD-L1

## Introduction

N-linked glycoproteins are widely expressed on the surface of immune cells and are closely correlated with immune responses and evasion [1]. Growing evidence reveal that the structures of *N*-glycans on specific proteins can regulate diverse protein interactions and biological functions, as well as modulate innate and adaptive immunity via sensing environmental signals. For example, abnormal branched *N*-glycans on immune checkpoint proteins could induce immunosuppressive networks in the tumor microenvironment through hidden immunosurveillance [2]. The in-depth study of *N*-glycan structures on specific glycoproteins and glycosites will drive an unprecedented development of glycobiology within the field of immunology [3]. Nevertheless, the structural diversity of *N*-glycans poses analytical challenges that limit the exploration of specific glycosylation functions [4].

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Over the past decades, MS (mass spectrometry)-based proteomics provided great opportunities to analyze glycoproteins or de-glycosylated peptides [5]. Previous studies have coupled MS analysis with de-glycosylated peptides after *N*-glycan release to enable the comprehensive analysis of glycoproteins and their *N*-glycosites, in which the glycan information was completely lost [6,7]. To overcome these shortcomings, intact glycopeptide analysis has been used to identify glycoproteins with site-specific glycan information in biological samples [8]. Unfortunately, these approaches and tools for direct intact glycopeptide analysis neither identify the functional glycan structures nor distinguish the different isomeric structures. To overcome these limitations, we recently developed a new software tool called StrucGP, which is based on a glycan database-independent method for structural interpretation of glycosite-specific glycans at the proteomic level and enables us to distinguish different structural isoforms within the same *N*-glycan compositions and identify new *N*-glycan structures [9]. So far, StrucGP has been successfully applied to many research fields, such as male reproduction and cancers [10,11].

Macrophages can be differentiated into classically activated macrophages (pro-inflammatory M1) and alternatively activated macrophages (anti-inflammatory M2), which play important roles in different physiologic and pathological processes [12]. In recent years, MS-based glycomics and glycoproteomics have been widely used for macrophage studies. Zarif *et al.* [13] coupled the hydrazide chemistry-based solid-phase capture of glycopeptides method with the use of PNGase F for the specific release of formerly *N*-glycosylated peptides and performed LC-MS/MS analysis to obtain high throughput identification and quantification of glycoproteins in primary human macrophages. Hinneburg *et al.* [14] employed label-free porous graphitized carbon–liquid chromatography-tandem MS to profile the *N*- and *O*-glycome associated with primary human monocyte-to-macrophage transition and found that monocytes and macrophages displayed similar *N*-glycome profiles. However, the low cell number and heterogeneity of macrophages obtained from donors often represent a barrier in analyzing the functions of primary cells; thus, human THP-1 monocyte-derived macrophages with similarities to primary macrophages have been widely used to circumvent these problems [15]. Delannoy *et al.* [16] investigated the alterations of N- and O-linked glycans during the differentiation of monocytic THP-1 cells into macrophages by using a simple and rapid method for the permethylation of carbohydrates. Their study revealed that macrophage differentiation increased the complex-type structures and slightly decreased the proportion of multifucosylated *N*-glycans and $\alpha$2,6-sialylation. Kalxdorf *et al.* [17] used an isobaric mass tag-based chemical labeling strategy that enabled the time-resolved analysis of plasma membrane protein presentation during the differentiation of the THP-1 monocytes into macrophages. In addition, by combining metabolic labeling, biorthogonal chemistry, and multiplexed MS-based proteomics, Suttapitugsakul *et al.* [18] quantified the dynamics of surface glycoproteins in THP-1 monocytes and macrophages comprehensively and site-specifically in response to lipopolysaccharide (LPS). These studies have contributed substantially to our understanding of the importance of glycosylation on macrophages, whereas none of them obtained the corresponding information between glycan structures and glycosites.

In this study, by applying the StrucGP method to *N*-glycoproteomic analysis of human THP-1-derived macrophages (M0, M1, and M2), significant changes in *N*-glycan structures, as well as their glycoproteins, were detected in different macrophages in addition to the identification of a large number of intact glycopeptides. These data provided available references for the further study of the site-specific *N*-glycans in macrophages, as well as the preclinical studies using the THP-1-derived macrophage model. More importantly, this study revealed the great potential of the StrucGP in immunological research.

## Materials and methods

### Cell culture and stimulation

Human monocytic cell line (THP-1) was obtained from the cell bank of the Chinese Academy of Sciences (Shanghai, China). THP-1 monocytes were cultured in RPMI 1640 medium (HyClone, USA), supplemented with 10% fetal bovine serum (Biolnd, Israel; VivaCell, Shanghai, China) and 1% penicillin–streptomycin (Solarbio, China) at 37 °C and 5% $CO_2$. The THP-1 monocytes were differentiated to macrophages with 10 ng/mL phorbol-12-myristate-13-acetate (PMA) for 24 h. Subsequently, the THP-1 macrophages were stimulated with human recombinant human interferon-γ (50 ng/mL) and LPS (15 ng/mL) for 48 h to M1 phenotype or stimulated with human IL-4 (25 ng/mL) and IL-13 (25 ng/mL) for 72 h to M2 phenotype. The untreated PMA-THP-1s were used as M0 phenotype. THP-1 cells-derived M0, M1, and M2 macrophage samples were harvested as described previously, and the successful polarization of M1 and M2 macrophages had been confirmed, as described previously [19].

### Cell lysis and protein extraction

The methods for cell lysis and protein extraction have been described in details in a previous publication [19]. Specifically, cells were washed three times by ice-cold PBS buffer to remove the cell culture medium. The denaturing buffer that contained 8 M urea and 1 M ammonium bicarbonate was added to each cell culture dish for cell lysis. To reduce the biological variances, cellular proteins were harvested from three biological replicates at each condition, and the proteins were pooled into one sample for further sample preparation. Briefly, denatured proteins in denaturing buffer were reduced by 5 mM dithiothreitol (DTT) at 37 °C for 1 h and alkylated by 15 mM iodoacetamide (IAM) at room temperature in the dark for 30 min. Reaction was terminated by 2.5 mM additional DTT at room temperature for 10 min. The solution was diluted two times with ultra-pure water, and then the proteins were digested by sequencing grade trypsin (Promega, Madison, WI, USA) with the ratio of 1:100 (trypsin to total protein, w/w) at 37 °C for 2 h. The solution was further diluted four times with ultra-pure water, and additional trypsin (trypsin to total protein, 1:100, w/w) was added with overnight incubation at 37 °C overnight. After protein digestion, the pH of the solution was adjusted with 10% trifluoroacetic acid

(TFA) until the pH < 2. The sample solution was centrifuged at 15 000 g for 10 min, and the peptides in the supernatant were desalted using hydrophile–lipophile balance (HLB) columns (Waters, Milford, MA, USA). The peptides were eluted from the column by 60% acetonitrile (ACN)/0.1% TFA.

**Enrichment of glycopeptides**

The intact glycopeptides were enriched using hydrophilic interaction chromatography (HILIC) micro-column (Agela Technologies, Tianjin, China), and the method has been described in detail in previous publications [9,20]. Briefly, one aliquot of the tryptic peptides eluted from the HLB column was diluted to a final solvent composition of 80% ACN/1% TFA for HILIC enrichment. Before sample loading, the HILIC micro-columns were washed twice each by 0.1% TFA and 80% ACN/0.2% TFA. After loading the samples, the columns were washed three times with 80% ACN/0.2% TFA. The glycopeptides bound to the column were eluted in 0.1% formic acid (FA) solution.

**MS analysis**

Each sample underwent three LC–MS/MS runs on an Orbitrap Fusion Lumos mass spectrometer (Thermo Fisher Scientific, Germany). Approximately 1 μg of glycopeptides was separated by an Easy-nLC 1200 system with a 75 μm × 50 cm Acclaim PepMap-100 C18 separating column protected by a 75 μm × 2 cm guarding column. The mobile phase flow rate was 200 nL/min and consisted of 0.1% FA in water (A) and 80% ACN/0.1% FA (B). The gradient profile for LC separation was set as follows: 3%–40% B for 203 min, 40%–68% B for 20 min, 68%–99% B for 4 min, and 99% B for 13 min. The spray voltage (+) was set at 2400 V. Orbitrap spectra with automatic gain control (AGC) of $4 \times 10^5$ were collected from 375 to 2000 *m/z* at a resolution of 120 000 followed by oxonium ions (138.055 *m/z* and 204.087 *m/z*)-dependent HCD (33% collision energy) triggered 20% collision energy HCD MS/MS at a resolution of 30 000 using an isolation width of 2 *m/z*. Charge states from 2 to 7 were selected for MS/MS acquisition. Unassigned and singly charged ions were rejected. A dynamic exclusion time of 20 s was used to discriminate against previously selected ions.

**Identification of intact glycopeptides by StrucGP**

The identification of intact glycopeptides was performed by StrucGP. Briefly, all MS data were first converted into "mzML" format by the Trans-Proteomic Pipeline (TPP, v.5.0.0) [21]. As previously described [9], the intact glycopeptide analyses were performed using the built-in glycan branch structure database from StrucGP and the human protein databases (UP000005640). The protein enzymatic digestion was set as trypsin with a maximum of two missed cleavage sites, and the potential glycosite-containing peptides were screened with the N-X-S/T motif (X is any amino acid except Proline). The carbamidomethylation (C, +57.02 Da) was set as a fixed modification and oxidization (M, +15.9949 Da) as a dynamic modification. The mass tolerances for MS1 and MS2 were set at 10 ppm and 20 ppm, respectively. To determine the Y ions, an optional mass shift of ± 1 or ± 2 Da was allowed in addition to the 20-ppm mass tolerance in MS2. The peptide sequences were determined by b+/y+ ions in the spectra of high HCD energy (HCD = 33%). Under low HCD energy (HCD = 20%), the glycan subtypes and core structures can be identified using mostly Y ion patterns, whereas the branch structures can be interpreted by a combined use of feature Y and B ions. Finally, the peptide and glycan portions required < 1% false discovery rates (FDR) for the intact glycopeptide identification. The principle of the method and the algorithms used have been described in previous publications [9,22].

**Structural mimicry of a protein model and visualization of the results of StrucGP**

The structural mimicry of the PD-L1 model was made by UCSF ChimeraX 1.2 [23]. GlycoVisual tool was used to visualize the results of StrucGP to display the corresponding MS/MS spectra automatically. These MS/MS spectra could demonstrate the principle of glycan type and detailed structure determination in intact glycopeptide analysis. In addition, the function of showing all glycan structures on a given glycopeptide greatly simplified the data analysis process. StrucGP and GlycoVisual Tool can be downloaded from the Zenodo database.

**Bioinformatics analysis**

Gene ontology (GO), Kyoto Encyclopedia of Genes and Genomes (KEGG), Reactome pathways, and protein interactions were analyzed using the STRING database [24]. The gene names of proteins were inputted into the STRING database with species set to human, and GO, KEGG and Reactome pathways analyses by whole genome as background. In addition, the STRING database can provide a critical assessment, and the integration of protein–protein interaction (PPI) with a combined score of > 0.4 was used to assess the direct (physical) and indirect (functional) associations of unique glycoproteins of M1 or M2 macrophages. GO analysis was undertaken to facilitate the understanding of the unique biological significance of the glycoproteins. The pathway analysis of glycoproteins was carried out to

determine the important pathways based on the KEGG and Reactome pathways.

## Results

### Profiling overview of site-specific *N*-glycosylation in THP-1-derived macrophages

The intact glycopeptides from M0, M1, and M2 macrophages were enriched and analyzed using MS-based glycoproteomic approaches, followed by the large-scale mapping of site-specific glycan structures via StrucGP (Fig. 1A). The glycosites and their corresponding glycan distributions were summarized and exhibited according to the color intersection in the heat map. A total of 2920 intact glycopeptides, which comprised 253 *N*-glycans with distinct structures and 652 glycosites, were identified from 372 *N*-glycoproteins in all three macrophage subtypes (Fig. 1B, Table S1). Specifically, 135 (451), 163 (380), and 201 (424) *N*-glycan structures (glycosites) were attached to 1438 (264), 1441 (221), and 1848 (240) unique *N*-glycopeptides (glycoproteins) in M0, M1, and M2 macrophages (Fig. 1C, Table S2).

### StrucGP reveals glycan isoforms in THP-1-derived macrophages

StrucGP can distinguish different glycan isoforms because of its glycan structure interpretation ability. Our data showed that up to four distinct glycan isoforms could be discriminated from one single composition in THP-1-derived macrophages, and the numbers of *N*-glycans that contain distinct glycan isoforms were increased in M1 and M2 macrophages (Fig. 2). This StrucGP function obviously increases the depth of intact glycopeptide analysis.

### *N*-glycan structure interpretation of different macrophage subtypes

To assess the differences in glycosylation among three

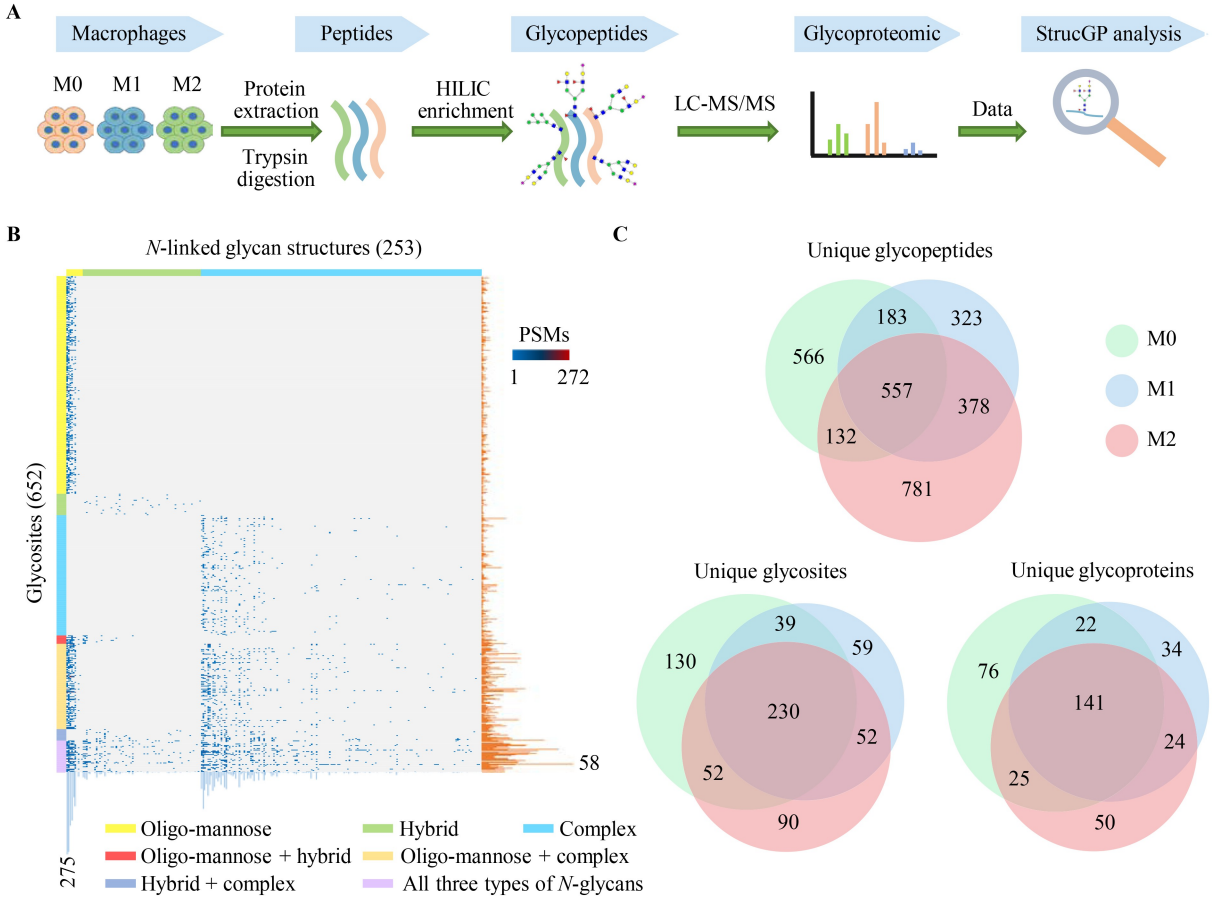



**Fig. 1** *N*-glycoproteomic analysis of THP-1-derived M0, M1, and M2 macrophages. (A) Workflow for the glycoproteomic analyses of three THP-1-derived macrophage subtypes. (B) Heat map of all identified intact glycopeptides in three THP-1-derived macrophage subtypes. The PSMs of intact glycopeptides, which comprises different glycan structures (upper) and glycosites (left), were shown in the heat map. The numbers of glycosites modified by each glycan and glycan numbers at each glycosite were summarized at the bottom (blue lines) and right (orange lines) parts of the figure, respectively. More details can be found in Table S1. (C) Venn diagram of all intact glycopeptides, glycosites, and glycoproteins identified in M0, M1, and M2 macrophages.

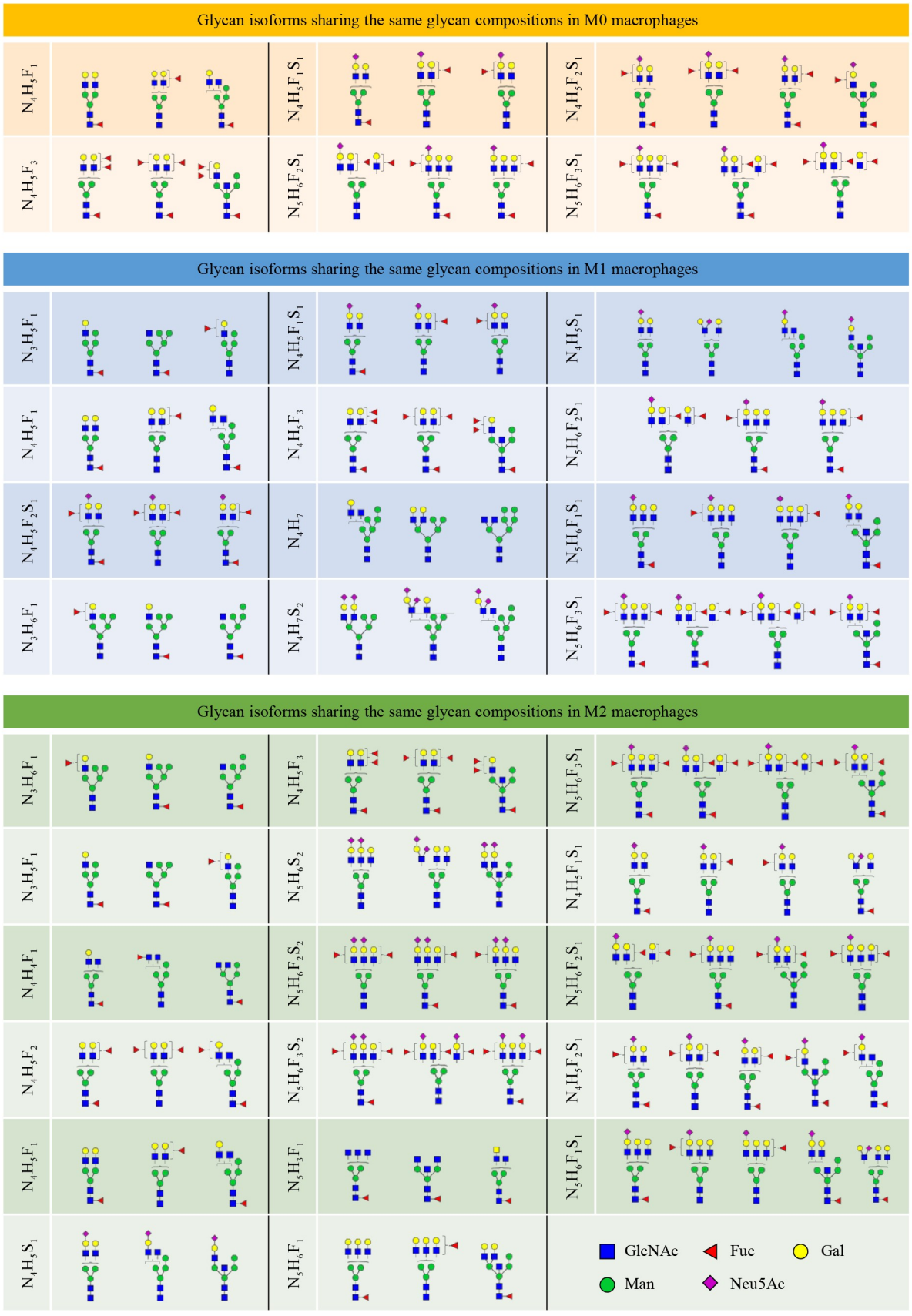



**Fig. 2** Glycan isoforms of THP-1-derived M0, M1, and M2 macrophages. *N*-glycan isoforms identified in M0, M1, and M2 macrophages. The location of each branch structure cannot actually be identified by StrucGP.

macrophage subtypes, the distributions of different *N*-glycans were analyzed in M0, M1, and M2 macrophages. Based on the numbers of unique glycopeptides, complex and hybrid glycans were only occupied by 27.0% and 3.43% in M0 macrophages but soared to 43.7% (60.7%) and 8.3% (8.2%) in the M1 (M2) macrophages, respectively (Fig. 3A). To explain the increase in the proportion of complex and hybrid glycans, we further analyzed *N*-glycans that contain fucose and/or sialic acid, and the results demonstrated that the proportions of *N*-glycans that contain fucose, sialic acid, and fucose + sialic acid were increased in M1 and M2 macrophages (Fig. 3B). Together, four types of core structure and the top ten branch structures were analyzed in three macrophage subtypes (Figs. 3C and 3D). For the core structure, the levels of core fucose and bisected GlcNAc were increased in M1 and M2 macrophages, especially in M2 macrophages. In addition, the proportions of branch structures that contain single HexNAc, LacNAc (HexNAc-Hex), and sialyl LacNAc (HexNAc-Hex-Neu5Ac) increase in M1 and M2 macrophages, whereas the proportion of oligo-mannose and Lewis epitopes ($HexNAc_1Hex_1Fuc_2$) decreased in M1 and M2 macrophages. Of note, based on the numbers of peptide-spectrum matches (PSMs), overall trends of altered *N*-glycans, core structures, and branch structures were consistent with the above results (Fig. S1).

## Analysis of bisected GlcNAc-modified glycoproteins in three macrophage subtypes

To explore the biological significance of the bisected GlcNAc structure in the macrophages, the glycoproteins modified by bisected GlcNAc with and without fucosylated core structure were listed in the different macrophage subtypes with *Z*-scored of PSMs (Fig. 4A). The three representative MS/MS spectra for the identification of intact glycopeptide from these glycoproteins were shown in Fig. S2. The results showed that the vast majority of the same glycoproteins were modified by more bisected GlcNAc in M2 than in M0 and M1. To determine the important pathways of bisected GlcNAc-modified glycoproteins and understand their biological significance in the macrophages, these proteins were further analyzed by GO, KEGG, and Reactome pathways (Fig. 4B). The results showed that the bisected GlcNAc-modified glycoproteins were mainly involved in multiple M2 macrophage-associated pathways, such as IL-4 and IL-13 signaling, regulation of the actin cytoskeleton, and receptor-mediated endocytosis. These findings indicate that bisecting *N*-glycans may be closely associated with M2 polarized macrophages.

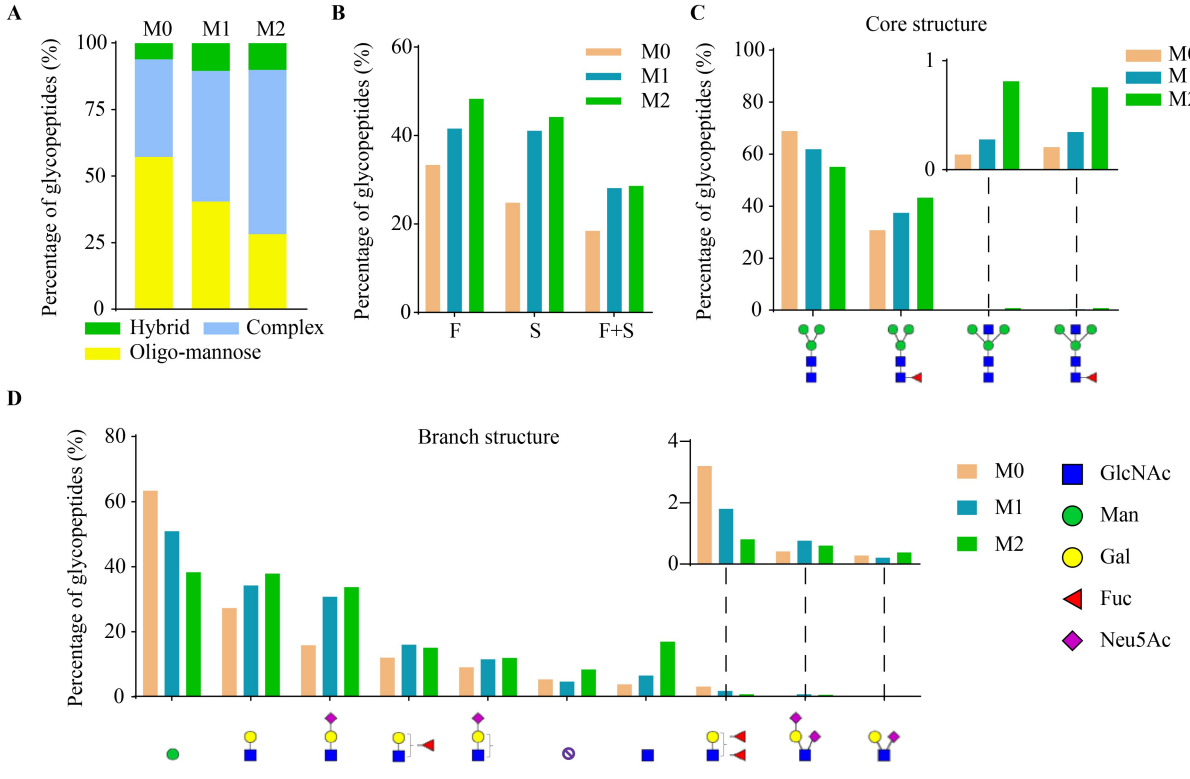



**Fig. 3** *N*-glycan structure interpretation of three human THP-1 macrophage subtypes. (A, B) Distributions of different glycan subtypes in various macrophage subtypes. (C, D) Distributions of four core structures and the top 10 branch structures (including the null Ø) identified in the different macrophage subtypes. The percentages in A–D were calculated on the basis of the numbers of unique glycopeptides. F: Fucose and S: Sialic acid.

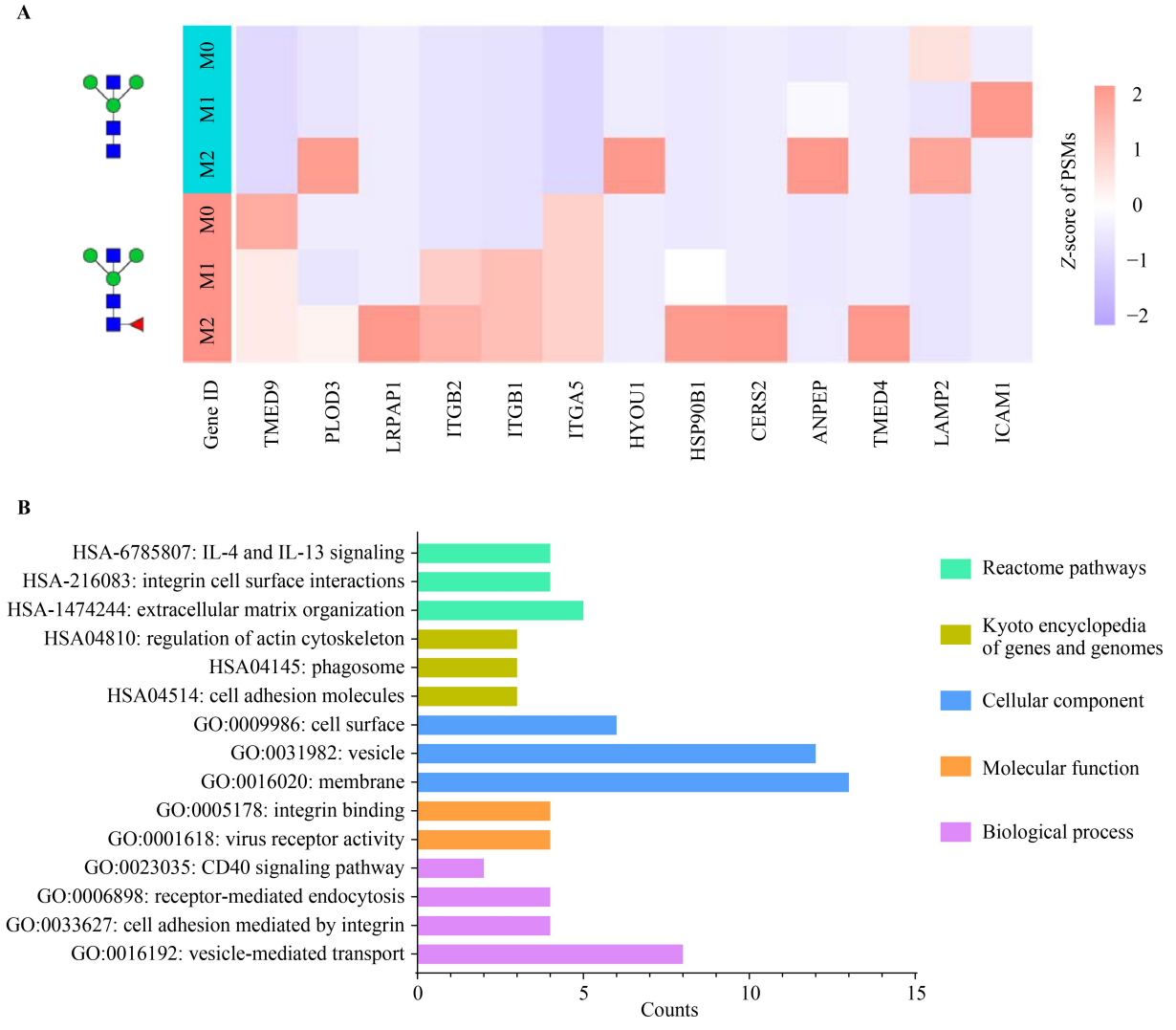
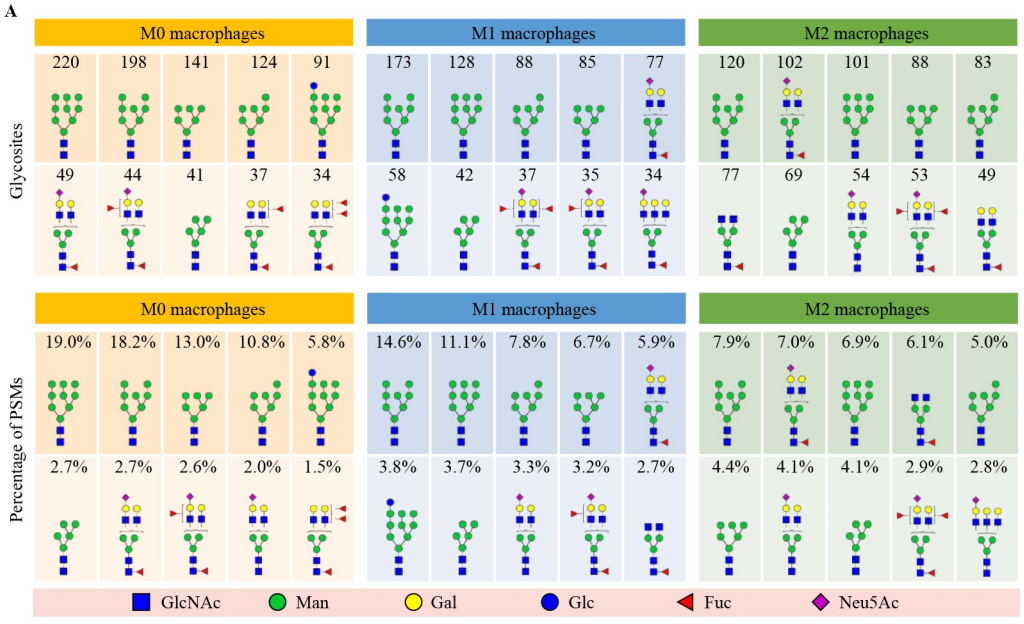
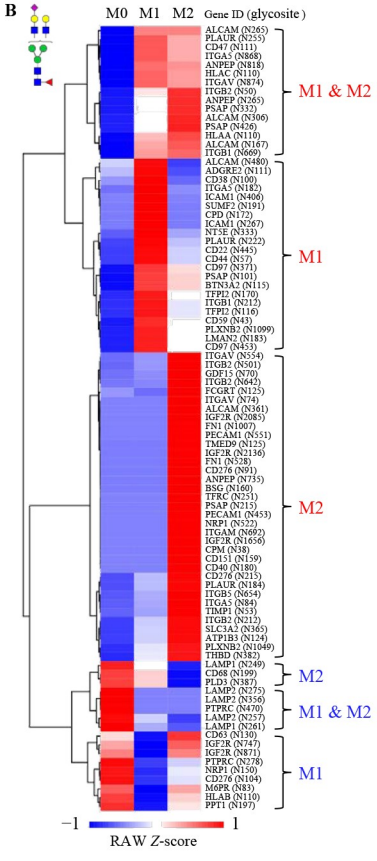



**Fig. 4** Analysis of bisected GlcNAc-modified glycoproteins in three macrophage subtypes. (A) Heat map of the bisected GlcNAc-modified glycoproteins in the different macrophage subtypes with *Z*-scored of PSMs. Each column corresponds to a protein gene ID. *Z*-score was measured on the basis of the standard deviations from the mean, and a high (low) *Z*-score represented a high (low) number of PSMs. The heat map was used only with the aim of ordering the PSMs of glycopeptides according to the column obtained *Z*-score ($z = (x - \text{mean})/\text{s.d.}$). (B) Biological function analysis of all bisected GlcNAc-modified glycoproteins in THP-1-derived macrophages. Enrichment *P* value < 0.05. The counts indicate how many proteins are annotated with a particular term. Digital labels of HSA (human sapiens) and GO (gene ontology) indicate the entries of annotation terms. Green represents the reactome pathways, yellow represents the Kyoto Encyclopedia of Genes and Genomes, blue represents the cellular component, orange represents the molecular function, and pink represents the biological process.

## Site-specific *N*-glycan structure analysis of different macrophage subtypes

To explore the biological significance of different *N*-glycan structures in macrophages, we analyzed the top 10 *N*-glycans detected on the glycoproteins of different macrophage subtypes based on the numbers of glycosites and PSMs (Fig. 5A). Interestingly, we found a representatively complex structure of $HexNAc_4Hex_5Fuc_1Neu5Ac_1$ ($N_4H_5F_1S_1$) in the top 10 *N*-glycans, which occupied more glycosites and accounted for many more PSMs in M1 and M2 macrophages compared with M0 macrophages. We further analyzed the glycopeptides (total PSMs ⩾ 2) modified by $N_4H_5F_1S_1$ in three macrophage subtypes and found that theses changed glycopeptides (glycoproteins) were mainly involved in multiple immune-associated processes, such as phagocytosis, inflammatory response, and PI3K-Akt signaling pathway (Fig. 5B). Regrettably, the relationship between biological functions and specific glycan structures is still unclear because of the limitations of technologies. Further studies are still needed to fully assess the effects of $N_4H_5F_1S_1$ glycan in macrophages.




| Types | Terms | Description | Counts | FDR (−lg) |
|---|---|---|---|---|
| M1 & M2 | GO:0002376 | Immune system process | 11 | 6.09 |
| | GO:0007166 | Cell surface receptor signaling pathway | 8 | 2.82 |
| | GO:0007229 | Integrin-mediated signaling pathway | 5 | 5.47 |
| | GO:0006909 | Phagocytosis | 4 | 2.82 |
| | HSA04151 | PI3K-Akt signaling pathway | 3 | 1.77 |
| | HSA-198933 | Immunoregulatory interactions between a lymphoid and a non-lymphoid cell | 4 | 2.96 |
| M1 | GO:0006954 | Inflammatory response | 4 | 1.63 |
| | GO:0007155 | Cell adhesion | 8 | 3.72 |
| | GO:0002376 | Immune system process | 9 | 2.24 |
| M2 | GO:0023035 | CD40 signaling pathway | 3 | 3.59 |
| | GO:0006909 | Phagocytosis | 5 | 2.34 |
| | GO:0042060 | Wound healing | 10 | 5.76 |
| | GO:0001525 | Angiogenesis | 7 | 3.6 |
| | GO:0006954 | Inflammatory response | 6 | 1.66 |
| | HSA04810 | Regulation of actin cytoskeleton | 7 | 5.04 |
| | HSA-6785807 | Interleukin-4 and interleukin-13 signaling | 5 | 2.8 |
| M2 | GO:0005765 | Lysosomal membrane | 3 | 2.24 |
| M1 & M2 | GO:0044754 | Autolysosome | 2 | 2.6 |
| M1 | GO:0032940 | Secretion by cell | 6 | 2.14 |
| | HSA04142 | Lysosome | 4 | 4.24 |
| | HSA04514 | Cell adhesion molecules | 3 | 2.36 |

**Fig. 5** Site-specific *N*-glycan structure analysis of different macrophage subtypes. (A) Top 10 *N*-glycans detected on the glycoproteins of different macrophage subtypes based on the numbers of glycosites and the numbers of PSMs. (B) Analysis of the $N_4H_5F_1S_1$-modified glycopeptides (total PSM ⩾ 2) in three macrophage subtypes. Cluster analysis of $N_4H_5F_1S_1$-modified glycopeptides in different macrophage subtypes with *Z*-scored of PSMs (left). *Z*-score was measured on the basis of the standard deviations from the mean, and a high (low) *Z*-score represented a high (low) number of PSMs. The heat map was used only with the aim of ordering the PSMs of glycopeptides according to the row obtained *Z*-score ($z = (x - \text{mean})/\text{s.d.}$). Red means upregulation, and blue represents the downregulation of expression. The biological function analysis of $N_4H_5F_1S_1$-modified glycoproteins in THP-1-derived macrophages (right). The counts indicate how many proteins are annotated with a particular term. Digital labels of HSA (human sapiens) and GO (gene ontology) indicate the entries of annotation terms. Enrichment *P* value < 0.05. The location of each branch structure (e.g., α2,3- and α2,6-linked sialic acids) cannot actually be identified by StrucGP.

### Differentially expressed site-specific *N*-glycans and glycoproteins in M1 and M2 macrophages

Next, we investigated the differences in *N*-glycan structures and glycoproteins that were uniquely identified between the M1 and M2 subtypes. Based on the numbers of PSMs, the analysis of the top five unique *N*-glycan structures in M1 and M2 macrophages showed that the *N*-glycan structures of M1 contained more Lewis epitopes and sialyl LacNAc; whereas the *N*-glycan structures of M2 contained more single HexNAc, bisected GlcNAc, and LacNAc (Fig. 6A). These special *N*-glycan structures are expected to be a distinction between the M1 and M2 macrophages.

In addition, we identified a list of glycoproteins that were uniquely found in each macrophage type, including 17 and 22 glycoproteins (PSMs ⩾ 2) that were uniquely identified in M1 and M2 macrophages, respectively. The M1-associated glycoproteins were mainly involved in defense response, innate immune response, INF-γ-mediated signaling pathway, and PD-1 signaling; meanwhile, the M2-associated glycoproteins were mainly associated with cell adhesion, regulation of the actin cytoskeleton, and integrin-mediated signaling pathway (Fig. 6B). Among the aforementioned glycoproteins, seven M1-related glycoproteins (including CD274, HLA-DRB1, HLA-DRA, IFI30, and PTGS2 as core glycoproteins with connections ⩾ 3), and nine M2-related glycoproteins (including LPL, CALR, FN1, and ITGA3 as core glycoproteins with connections ⩾ 3) had protein–protein interactions (Fig. 6C).

### *N*-glycan structures on PD-L1 expressed in M1 macrophages

Among the above core glycoproteins, a high expression

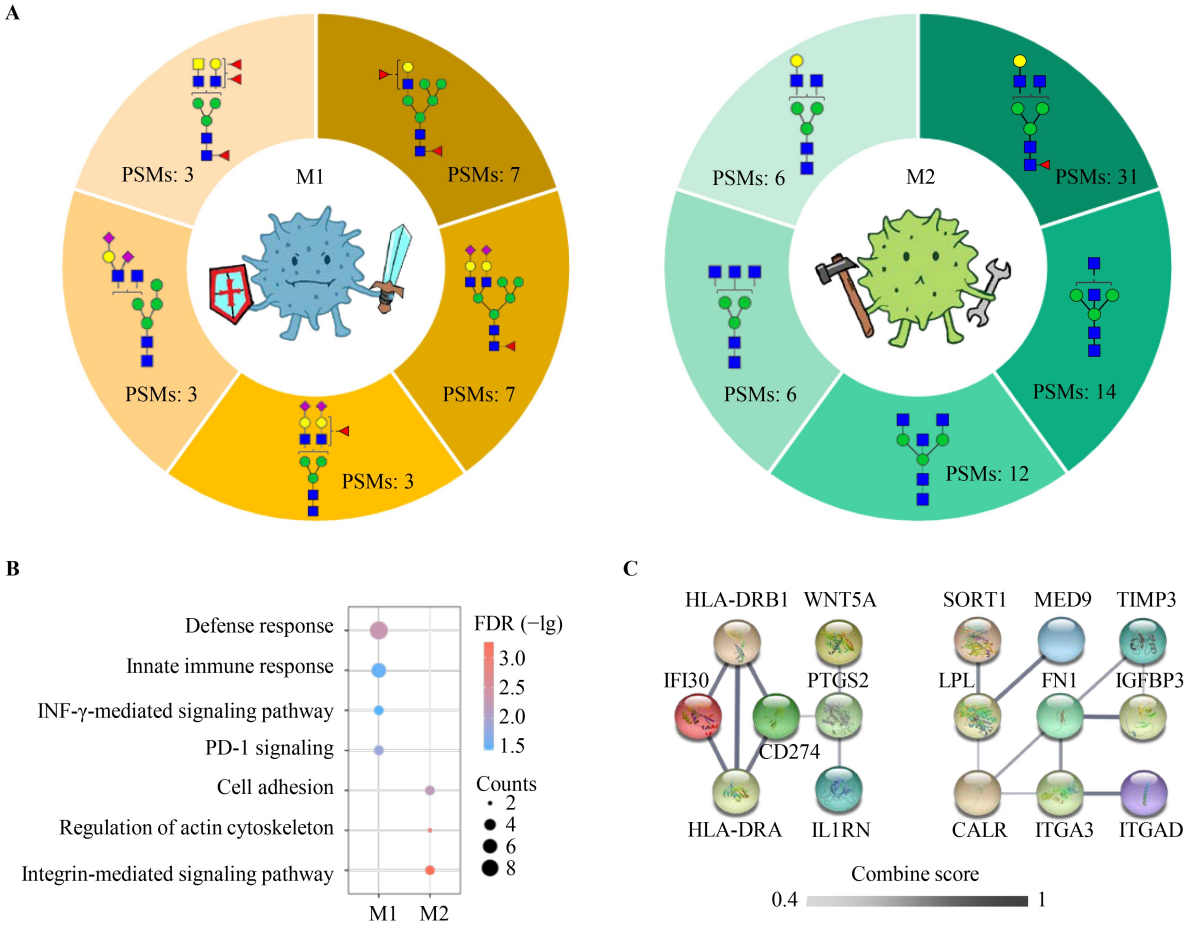



**Fig. 6** Differentially expressed site-specific *N*-glycans and glycoproteins in M1 and M2 macrophages. (A) Top five unique *N*-glycan structures in THP-1-derived M1 and M2 macrophages based on the numbers of PSMs. (B) Biological function analysis of unique glycoproteins in THP-1-derived M1 and M2 macrophages. The counts indicate how many proteins are annotated with a particular term. Enrichment *P* value < 0.05. (C) Protein–protein interaction (PPI) networks of highly interactive glycoproteins in M1 (left) and M2 (right) macrophages. The line colors varying from light to dark represent the combined score of proteins from 0.4 to 1. The location of each branch structure (e.g., α2,3- and α2,6-linked sialic acids) cannot actually be identified by StrucGP.

level of the immune checkpoint protein PD-L1 (CD274) was only detected in M1 macrophages [19]. By using StrucGP, two out of four potential *N*-glycosylation sites (e,g., Asn-192 and Asn-200) of PD-L1 modified by eight different glycans were solely identified in M1 macrophages in this study (Fig. 7A). Interestingly, the glycosite Asn-192 was mainly modified by high-mannose glycans, whereas the glycosite Asn-200 was occupied by three complex glycans with terminal sialylation and core fucosylation. In addition, the LacNAc moiety was detected at glycosites Asn-192 and Asn-200 of PD-L1, while two out of three glycans at the glycosite Asn-200 contained Lewis epitopes (e.g., $N_1H_1F_1$ and $N_1H_1F_1S_1$). Two representative MS/MS spectra (e.g., $N_3H_6$ and $N_4H_5F_1S_1$) were provided to demonstrate the principle of glycan type determination in the intact glycopeptide analysis (Fig. 7B), and others are shown in Fig. S3. These results suggest that the *N*-glycosylation of the same protein could vary with glycosites and macrophage subtypes.

## Discussion

*N*-glycosylation is an important post-translational modification of proteins that affect immune function, and the different patterns of glycosylation may lead to impaired protein expression, altered protein ligand function, and altered immune pathway signaling [25,26]. In recent years, *N*-glycans have been studied extensively in terms of their structures and precise functions, including the regulation of cytosolic and nuclear functions, inflammatory reactions, immune surveillance, hormone action, and tumoral immune escape [27]. The exploration of the function of glycoproteins in immune response has indicated the urgent need to analyze site-specific *N*-glycan on glycoprotein. To enable researchers to facilitate a broader exploration of the functional information content embedded within *N*-glycan structures, we developed a tool StrucGP for the structural interpretation of glycosite-specific *N*-glycans [9].

We demonstrated the potential of StrucGP in immunology research by using the *N*-glycoproteomic profiles of M0, M1, and M2 macrophages derived from THP-1 as examples. In the current study, we systematically characterized the glycoproteins with glycosites and attached glycan structures precisely in M0, M1, and M2 macrophages derived from THP-1 cells for the first time. StrucGP could distinguish many different structural isomers within the same *N*-glycan composition. Previous studies showed that the different glycan isomers play key roles in the development and progression of many diseases while various approaches have been developed to facilitate the characterization of glycan isomers [28]. A recent study showed that the differential recognition of oligo-mannose isomers by glycan binding proteins involved in innate and adaptive immunity could affect the protein–glycan interactions [29]. This finding showed the importance of glycan isomers for immune-glycobiology studies. However, StrucGP only showed the glycan isoforms of THP-1-derived macrophages in this study, the significance of different isomers to immune response will be further explored in future studies.

In this study, our results showed that bisected GlcNAc and core fucosylated and sialylated glycans were increased in M1 and M2 macrophages, especially in the latter. In a previous study, Yang *et al.* [30] applied a lectin array to study the glycosylation of the THP-1-derived macrophages during polarization, and our findings from the data set substantiate that core fucose (recognized by *Pisum sativum* agglutinin and *Lens culinaris* agglutinin), bisected GlcNAc (recognized by *Phaseolus vularis* erythroagglutinin), and sialic acid (recognized by *Maackia amurensis*, *Maackia amurensis* lectin I, *Sambucus nigra*, and *Sambucus nigra* agglutinin I) were increased in M1 and M2 macrophages, and a significant increase was shown in the latter. This consistency further confirms the reliability and generality of StrucGP in *N*-glycoproteomic analysis. Recent evidence suggests that terminal fucosylation is a hallmark of inflammatory macrophages and might contribute to the commitment, differentiation, and maturation of M1 inflammatory macrophages [31,32]. In addition, Zhao *et al.* confirmed that sialylation status is closely related to macrophage biomechanical characteristics (e.g., elastic modulus, tether force, tether radius, adhesion force, and membrane tension) and directly involved in macrophage function [33]. However, no direct evidence proves the association between bisected GlcNAc and macrophage polarization. Interestingly, we found that *N*-glycan $N_4H_5F_1S_1$ occupied more glycosites and accounted for many more PSMs in M1 and M2 macrophages than in M0 macrophages and is closely related to the immune-associated processes of macrophages. Unfortunately, we are currently unable to evaluate the functions of $N_4H_5F_1S_1$ in macrophages because of technical limitations. Nevertheless, StrucGP provides a foreshadowing for the future research function of site-specific *N*-glycan structures on glycoproteins through the structural analysis of *N*-glycans. The investigation of $N_4H_5F_1S_1$ may also be undertaken at a later date.

The most important finding of this study was that the high expression of glycosylated PD-L1 was identified in M1 macrophages. In a recent study, Cai *et al.* reported that traditionally defined M1 macrophages with "anti-tumor properties" could facilitate cancer progression through high PD-L1 expression [34]. As an important cell surface glycoprotein, PD-L1 would be degraded by the ubiquitin/proteasome system when the protein was not

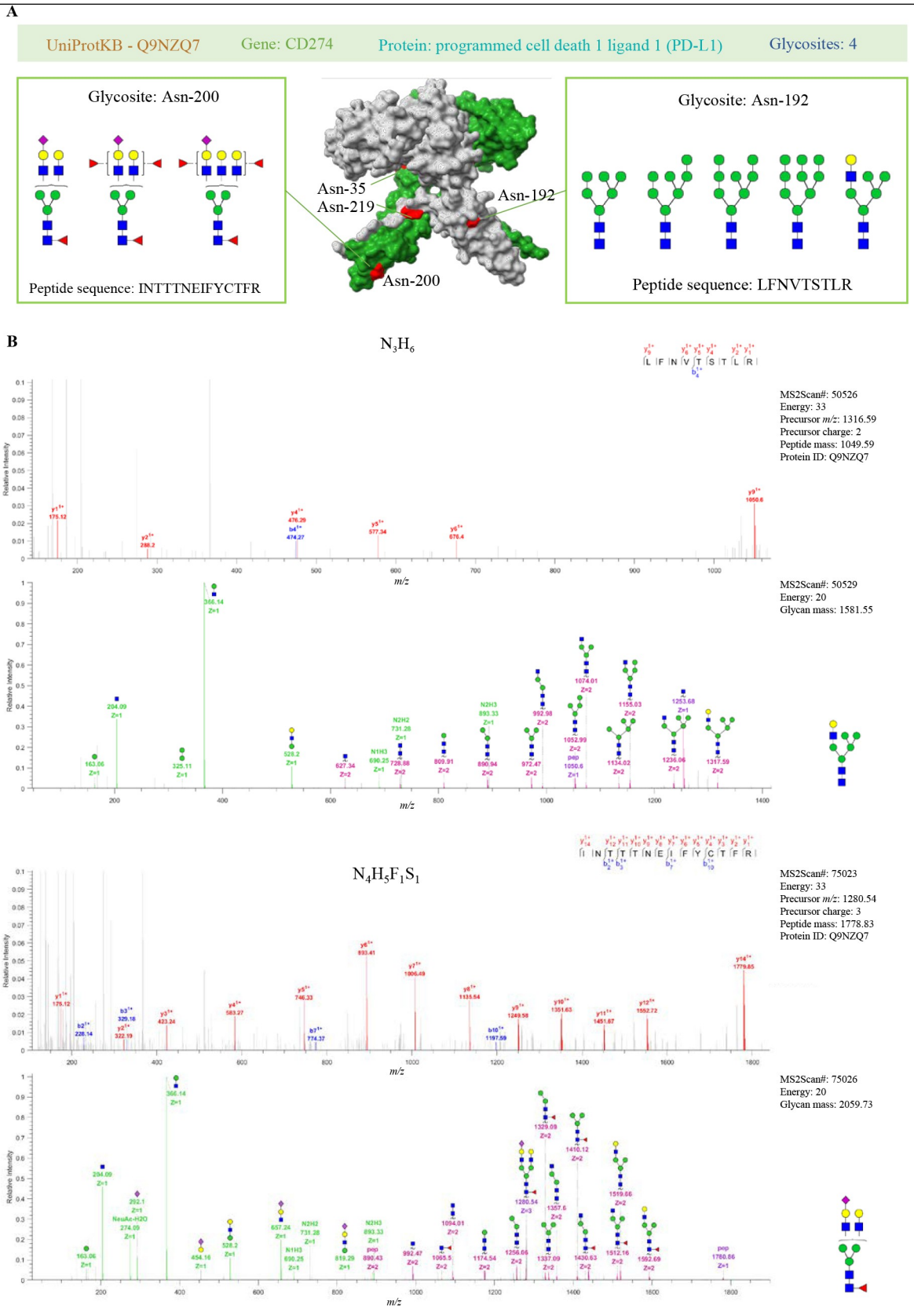



**Fig. 7** *N*-glycan structures of PD-L1 protein in THP-1-derived M1 macrophages. (A) *N*-glycan structures on glycosites Asn-192 and Asn-200 of PD-L1. (B) Representative MS/MS spectra for the identification of an intact glycopeptide from PD-L1. Upper: peptide sequence identification using a MS/MS spectrum of high HCD energy (HCD = 33%). Lower: glycan structure determination using MS/MS spectra of low HCD energy (HCD = 20%). The location of each branch structure (such as α2,3- and α2,6-linked sialic acids) cannot actually be identified by StrucGP.

glycosylated, and the stability of protein was proven to be mainly maintained by the glycosylation at glycosites Asn-192, Asn-200, and Asn-219 but not Asn-35 [35]. In addition, previous reports showed that the glycan structures on Asn-192 and Asn-200 of human PD-L1 contained poly-*N*-acetyllactosamine (poly-LacNAc), which is catalyzed by β-1,3-*N*-acetylglucosaminyl transferase (B3GNT3), thereby mediating PD-L1 and PD-1 interaction [36]. In our study, StrucGP allowed us to further reveal the distinct structural profiles on site-specific *N*-glycans of PD-L1. In a nutshell, the recommendations and future perspectives of this work include targeting glycosylated PD-L1, which could make up for the defect in tumor-associated macrophages and may be a potentially promising cancer treatment strategy.

Precision structural interpretation of site-specific glycans, as well as the subsequent intervention of target glycoproteins and related glycosyltransferases, are of high value for the development of new diagnostic and therapeutic approaches for several diseases. In this work, we only used THP-1-derived macrophages as an immune cell model example to show the value of StrucGP in immunology research. Notably, the *N*-glycosylation of the same glycoprotein may vary in different types of cells, and further research in immune cells of different sources is encouraged to be undertaken.

In addition, StrucGP is unable to distinguish the left/right branches of the *N*-glycan structures, and the structural interpretation of fucose-containing glycans might be influenced by the rearrangement of the fucose residues in these glycans during mass spectrometry-based fragmentation. Moreover, even though mixed spectra are widespread, StrucGP can only identify one isomer per spectrum. Given these limitations, additional evidence may still be required for the further validation of these structures. Finally, although StrucGP still has to be developed, it has the potential to be a new site-specific *N*-glycan analysis tool in immunological research.

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## Compliance with ethics guidelines

Pengfei Li, Zexuan Chen, Shanshan You, Yintai Xu, Zhifang Hao, Didi Liu, Jiechen Shen, Bojing Zhu, Wei Dan, and Shisheng Sun declare that they have no conflict of interest. All authors read and approved the final manuscript and, therefore, had full access to all the data in the study and take responsibility for the integrity and security of data. This article does not contain any studies involving human or animal subjects.

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